Survival Analysis of Brazilian Companies: How Long Does a CNPJ Last and How Can 12-Month Closure Risk Be Anticipated?

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Abstract

This paper presents an applied study on firm longevity and short-term closure risk using Brazil's public CNPJ registry data. The project was structured as an auditable eight-phase pipeline, from raw-data consolidation to final reporting, around two central questions: how long Brazilian companies remain active and which registry signals help anticipate closure within 12 months. The survival layer was estimated on 67,175,202 firms and 31,342,895 observed closure events, with a Kaplan–Meier median of 14.17 years. The predictive layer was built with absolute temporal splits, explicit leakage control, and final model selection based on probabilistic quality. The chosen model, `logistic_full` calibrated with isotonic regression, achieved ROC-AUC of 0.6975, PR-AUC of 0.1327, Brier score of 0.0594, and a 2.53x lift in the top test decile. The main contribution is less a single algorithm and more a defensible methodological design for turning large administrative records into temporal inference and an operational risk score.

Keywords: firm survival; CNPJ; Kaplan–Meier; Cox; competing risks; calibration; predictive modeling.

1 Introduction

Estimating observed firm duration and anticipating closure signals is relevant for economic monitoring, risk products, and market intelligence. In massive administrative datasets, however, the main challenge is not only the final model. It begins with the definition of the analytical unit, the handling of time, the control of leakage, and the distinction between structural interpretation and operational prediction.

This project was designed to answer two complementary questions:

1. how long Brazilian firms last when observed through the public CNPJ registry;
2. which registry signals help anticipate closure over a 12-month horizon without violating temporal coherence.

The study uses the official CNPJ base layout released by Brazil's Receita Federal as its primary source [Receita Federal do Brasil, 2018]. The goal is not to deliver a causal explanation for firm mortality. The objective is to produce an auditable temporal reading of the registry and a probabilistic radar that is useful as an operational score.

2 Data and Study Design

2.1 Source and analytical unit

The CNPJ open data include variables such as registration number, registry status, status date, opening date, legal nature, and primary CNAE activity [Receita Federal do Brasil, 2018]. In this project, the main analytical unit was anchored at `cnpj_basico`, approximating the firm through the headquarters entity in order to keep origin, event, and temporal windows coherent.

The main survival event was defined as observed closure under `situacao_cadastral = 08`. Time origin was `data_inicio_atividade`, and administrative censoring was fixed at 2026-03-15. This makes it possible to interpret the problem as time to closure or censoring under a large registry snapshot.

2.2 Pipeline structure

The work was organized into eight phases: diagnosis and integrity, preprocessing, temporal auditing, exploratory analysis, survival, 12-month radar modeling, storytelling, and final artifacts. This decomposition matters because most methodological gain came from the sequence between data consistency, temporal design, and evaluation rather than from aggressive algorithmic tuning.

2.3 Temporal design of the 12-month radar

The closure radar was formulated as a probabilistic classification task with an absolute temporal split:

- training in 2022;
- validation in 2023;
- testing in 2024.

Features were deliberately kept compact and temporally defensible. Variables that could leak future information were removed during the audit phase, and final model selection prioritized calibration and stability in addition to discrimination.

3 Methodology

3.1 Survival layer

Survival curves were estimated with Kaplan–Meier [Kaplan and Meier, 1958]. As an additional explanatory layer, the project also fitted an exploratory Cox proportional hazards model [Cox, 1972]. This stage was treated as associative rather than causal because registry covariates represent an observed administrative snapshot rather than a complete longitudinal history.

3.2 Competing risks

Beyond overall survival, the project decomposed closure by administrative reason under a competing-risks perspective. The purpose was not to estimate causal mechanisms by cause,

but to separate the observed cumulative incidence of the main closure motives, in line with the competing-risks literature [Fine and Gray, 1999].

3.3 12-month risk radar

On the predictive side, several model families were compared under temporal validation. The final selection rule treated small PR-AUC differences as practical ties and broke ties through probabilistic quality, especially Brier score [Brier, 1950], log-loss, and operational simplicity. The chosen version was `logistic_full` with isotonic calibration.

4 Results

4.1 Overall survival profile

The final survival base contains 67,175,202 firms and 31,342,895 observed closure events, corresponding to an observed event rate of 46.66%. The Kaplan–Meier median survival time was 14.17 years.

Table 1: Estimated survival probabilities in the full sample

Horizon	Estimated survival
1 year	85.43%
3 years	75.74%
5 years	68.62%
10 years	57.18%

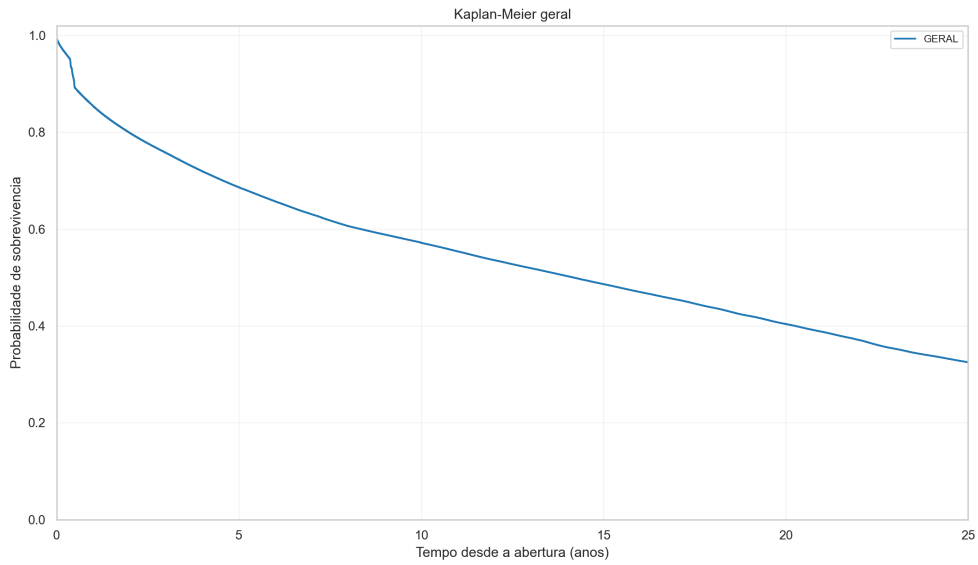

Figure 1: Overall Kaplan–Meier curve.

The pattern is consistent with concentrated risk in the first years of operation followed by a gradual slowdown. The correct reading here is structural and observational: the registry reveals the observed duration of registered firms, not a causal dynamic isolated from cohort composition.

4.2 Heterogeneity by cohort and size

The cohort curves show that opening period is one of the strongest interpretive dimensions in the project. Firms opened between 1995 and 1999 display an estimated 10-year survival of 80.63%, whereas the 2010–2014 cohort falls to 48.51%. Temporal composition is strong enough to reshape the interpretation of business duration.

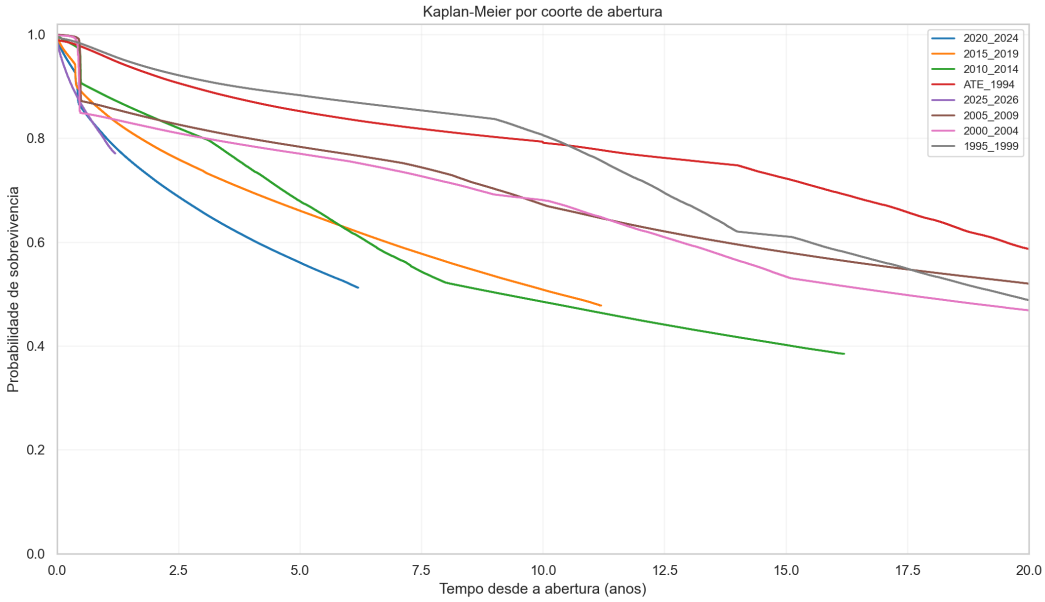

Figure 2: Kaplan–Meier by opening cohort.

Observed firm size also matters. Small companies classified as *empresa de pequeno porte* reach 81.38% survival at 10 years, while *micro empresa* firms remain at 55.45%. This should not be read as a simple causal effect of size. It shows that, when partitioned by administrative strata, the registry contains populations with sharply different survival profiles.

Table 2: Examples of heterogeneity at 10 years

Slice	Group	10-year survival
Cohort	1995–1999	80.63%
Cohort	2010–2014	48.51%
Size	Small company	81.38%
Size	Micro company	55.45%

4.3 Closure reasons and competing risks

The competing-risks layer shows that the main administrative closure cause at 10 years is voluntary liquidation, with cumulative incidence of 37.66%. The following causes are far smaller, such as registry cancellation at 3.00% and other motives at 0.55%.

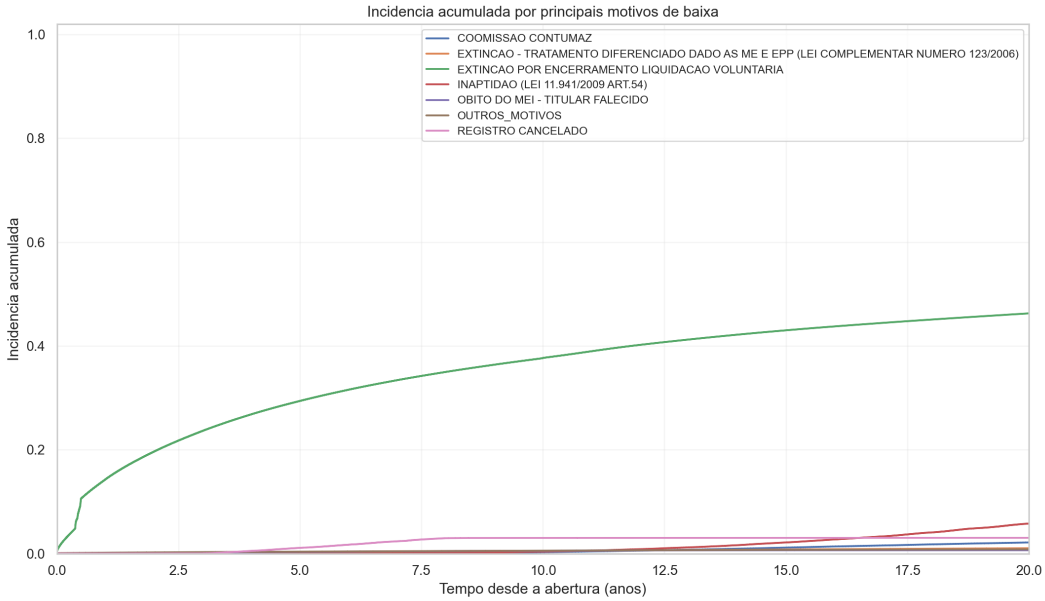

Figure 3: Cumulative incidence for the main closure reasons.

This decomposition is useful because it avoids collapsing all closure events into a single homogeneous block. At the same time, it remains an administrative reading of the registry. The economic meaning of each cause requires context beyond the base itself.

4.4 12-month closure-risk radar

On validation, random forest, XGBoost, LightGBM, and logistic regression performed very similarly in PR-AUC, all around 0.126. The final choice favored `logistic_full` because the practical tie in PR-AUC was broken by stronger probabilistic quality and simpler deployment.

On the 2024 temporal test split, the final model obtained:

- ROC-AUC of 0.6975;
- PR-AUC of 0.1327;
- Brier score of 0.0594;
- 2.53x lift in the top 10%;
- precision of 0.168 and recall of 0.247 at the high-risk threshold.

Table 3: Final model metrics on the test split

Metric	Value
ROC-AUC	0.6975
PR-AUC	0.1327
Brier	0.0594
Log-loss	0.2265
Lift top 10%	2.53x
High-risk threshold	0.1168
High-risk precision	0.1676
High-risk recall	0.2470

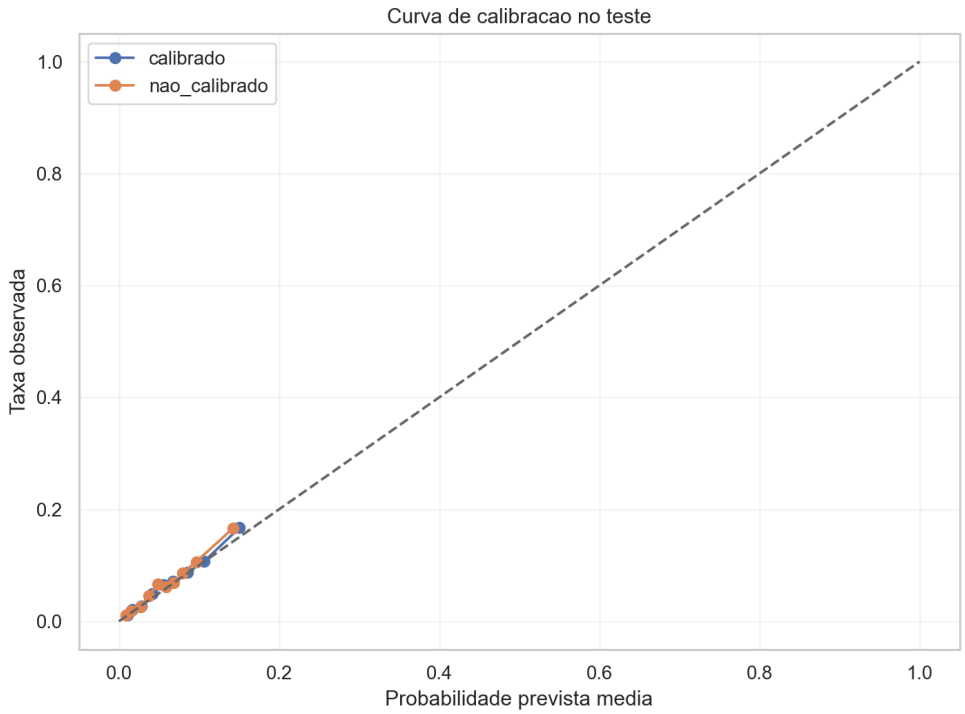

Figure 4: Score calibration on the temporal test split.

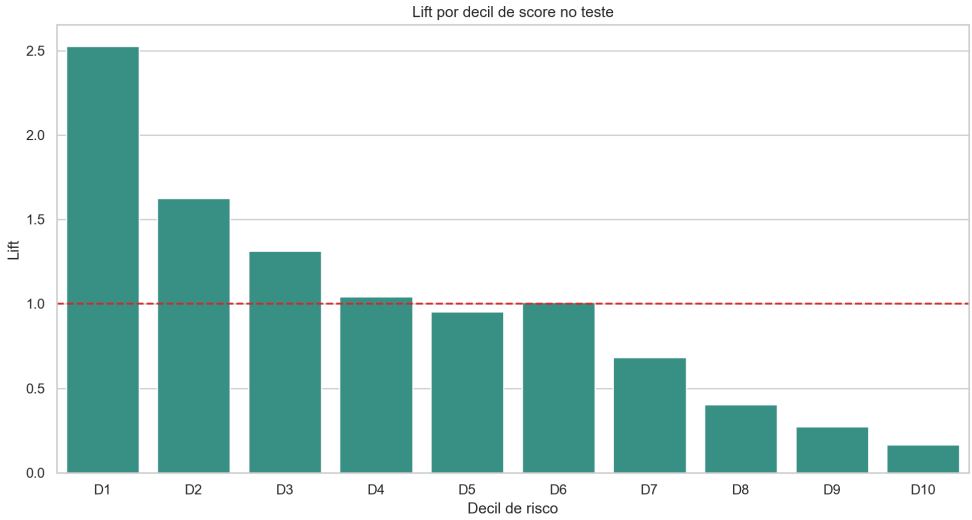

Figure 5: Decile lift on the temporal test split.

The operational reading matters more than any single metric. In a strongly imbalanced problem with a base event rate of 6.57%, concentrating 2.53 times the base rate in the top decile is a meaningful prioritization gain.

4.5 Strongest score signals

The strongest global signals in the final radar were firm age, Simples taxation status, history of exclusion from Simples, MEI status, and opening cohort. In practice, the score learns mostly from age and tax-regime structure rather than from a diffuse set of weak attributes.

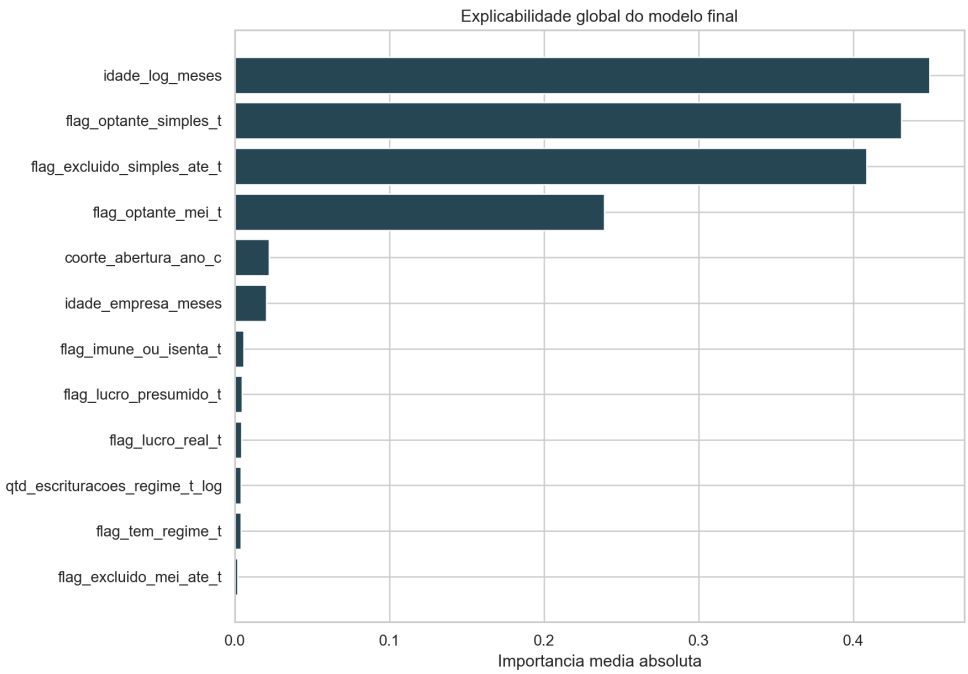

Figure 6: Global feature importance in the final radar.

This reinforces the methodological interpretation of the project: much of the radar's value comes from temporal discipline and conservative feature engineering rather than from algorithmic complexity for its own sake.

5 Discussion

The project delivers two complementary outputs.

The first is structural. The open CNPJ base allows a national-scale description of observed business duration, with a Kaplan–Meier median of 14.17 years and strong heterogeneity by cohort and firm size. Opening period proved especially important, suggesting that any aggregate reading without temporal stratification misses a central part of the story.

The second is operational. It is possible to build a methodologically defensible 12-month closure radar without future leakage and without relying on an overly complex model. The final score should not be treated as a causal explanation of firm failure. It is a probabilistic prioritization instrument under real administrative-data constraints.

The limitations are also clear. Part of the registry covariates is an observed administrative snapshot, not a full historical record. The Cox model was kept as exploratory. Some institutional groups require contextual interpretation. And, as in any large administrative base, the substantive meaning of a registry code does not necessarily map one-to-one onto the underlying economic mechanism.

6 Conclusion

This project shows that open administrative data can sustain a robust study of business survival and a useful 12-month closure-risk radar when temporal design, auditing, and methodological communication are treated as core parts of the solution.

More than producing a metric, the work organizes a complete analytical path: consolidated base, temporal inference, calibrated score, final figures, and technical article. Its main value lies in the combination of data scale, methodological discipline, and an honest reading of the limits of the evidence.

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